

Batocera v42 Exhaust Fans Setup

Setup & Troubleshooting Reference

Raspberry Pi 5 5 GPIO Pin (Physical) 4, 9, 32, 36

Connector wiring - 4 wires only

Fan Cables	Pi 5 Physical Pin	Function
Both Fan Red Wires twisted together	Pin 4	5V Power
Both Fan Black Wires twisted together	Pin 9	GND
Both Fan Blue Wires twisted together	Pin 32	PWM Signal (Currently script is not using PWM, but still keep it connected for future improvements)
Fan 1 Yellow Wire only	Pin 36	Tachometer (fan speed)

Why PWM was abandoned

Hardware PWM overlays for Pi 5 RP1 chip not supported in Batocera's kernel. Software PWM via subprocess caused stuttering due to process launch overhead. Busy-wait PWM loop consumed one full CPU core at 100%. Longer cable fans stuttered due to signal degradation at high PWM frequencies. Final decision - simple on/off control via `lgpio gpio_write`. Zero CPU overhead, both fans behave identically, clean signal.

Temperature thresholds

- Fans ON above 65°C
- Fans OFF below 58°C (thresholds can be changed in the script)
- 5-7°C hysteresis gap prevents rapid on/off cycling at threshold boundary
- Pi 5 throttles at 80°C, rated safe to 85°C

1. Batocera Configuration

1.1 Create script fan_pwm.py:

Create this file on Batocera Alt + F5 or via SSH:

```
| nano /userdata/system/scripts/fan_pwm.py
```

Paste these code inside it:

```
#!/usr/bin/env python3

import lgpio
import time
import os
import signal
import sys
from datetime import datetime

GPIO_CHIP      = 0
GPIO_PWM       = 12
GPIO_TACH      = 16
CHECK_INTERVAL = 10
LOG_FILE       = "/userdata/system/logs/fan_control.log"

TEMP_ON  = 65
TEMP_OFF = 58

os.makedirs(os.path.dirname(LOG_FILE), exist_ok=True)

def log(msg):
    timestamp = datetime.now().strftime("%Y-%m-%d %H:%M:%S")
    line = f"{timestamp} {msg}"
    print(line)
    with open(LOG_FILE, "a") as f:
        f.write(line + "\n")

def get_temp():
    with open("/sys/class/thermal/thermal_zone0/temp") as f:
        return int(f.read().strip()) // 1000

h = lgpio.gpiochip_open(GPIO_CHIP)
lgpio.gpio_claim_output(h, GPIO_PWM)
lgpio.gpio_claim_input(h, GPIO_TACH)

def cleanup(signum=None, frame=None):
    lgpio.gpio_write(h, GPIO_PWM, 0)
    lgpio.gpiochip_close(h)
    log("Fan control stopped - fans off")
    sys.exit(0)

signal.signal(signal.SIGTERM, cleanup)
```

```

signal.signal(signal.SIGINT, cleanup)

log("Fan control started")
log(f"Fan ON above {TEMP_ON}C | Fan OFF below {TEMP_OFF}C")

fans_on = False

while True:
    temp = get_temp()

    if not fans_on and temp >= TEMP_ON:
        lgpio.gpio_write(h, GPIO_PWM, 1)
        fans_on = True
        log(f"Temp: {temp}C | Fans ON")

    elif fans_on and temp <= TEMP_OFF:
        lgpio.gpio_write(h, GPIO_PWM, 0)
        fans_on = False
        log(f"Temp: {temp}C | Fans OFF")

    time.sleep(CHECK_INTERVAL)

```

Save with Ctrl+O, exit with Ctrl+X.

1.2 Use **custom.sh** for script auto start:

Use **custom.sh** which is Batocera's official autostart hook:

```
| nano /userdata/system/custom.sh
```

Replace this in the entire file (volume button, fan control script auto start)

```

#!/bin/bash
modprobe fuse
bash "/userdata/system/configs/firefox/restore_desktop_entry.sh" &
python3 /userdata/system/scripts/volume_buttons.py &
python3 /userdata/system/scripts/fan_pwm.py &

```

Or Add this to the bottom:

```

#!/bin/bash
python3 /userdata/system/scripts/fan_pwm.py

```

Save with Ctrl+X → Y → Enter, then:

```
| chmod +x /userdata/system/custom.sh
```

1.3 Manually test / control fans:

Start script:

```
| python3 /userdata/system/scripts/fan_pwm.py &
```

Stop script and turn fans off:

```
| pkill -f fan_pwm.py  
python3 -c "import lgpio; h=lgpio.gpiochip_open(0);  
lgpio.gpio_claim_output(h,12); lgpio.gpio_write(h,12,0);  
lgpio.gpiochip_close(h) "
```

Check running processes:

```
| ps aux | grep fan
```

Check log:

```
| cat /userdata/system/logs/fan_control.log
```

Check temperature:

```
| watch -n 5 vcgencmd measure_temp
```

Reboot:

```
| reboot
```

After reboot check for logs to see if script auto started or not

```
| ps aux | grep fan  
cat /userdata/system/logs/fan_control.log
```

2. Problems Faced

- PWM overlay isn't supported properly by the Batocera kernel and the internal HAT fan was interfering with the GPIO PWM signal causing the external fans to stutter, the one with longer cable was not even starting.
- So skipped PWM for now, the fans run at full speed when a pre-set temperature threshold is crossed i.e. above 65 degrees Celsius and stops when temperature drops below 58 degrees Celsius. (threshold can be changed in the script)